CLASS 10 - SCIENCE

AI-Assisted Evaluation Demo Examination

Time: 45 minutes Maximum Marks: 25

Instructions

1. Attempt all questions.
2. Show calculation steps wherever applicable.
3. Write answers clearly and label each answer with the correct question number.
4. Use SI units wherever required.

Q1. Explain why a metal spoon feels colder than a wooden spoon when both are kept in the same room for a long time. **[3 marks]**

Q2. A 12 V battery is connected across a 6 ohm resistor. Calculate (a) the current through the resistor and (b) the electrical power consumed. **[4 marks]**

Q3. State two differences between aerobic respiration and anaerobic respiration, and give one example of each. **[3 marks]**

Q4. A student places an iron nail in copper sulphate solution. Describe two observations and write the balanced chemical equation for the reaction. **[4 marks]**

Q5. Why does the sky appear blue during the day? Explain using the scattering of light. **[3 marks]**

Q6. A plant is kept in the dark for 24 hours and then one leaf is partly covered with black paper before placing the plant in sunlight. After a few hours, the leaf is tested with iodine. What result is expected and what does the experiment show? **[4 marks]**

Q7. Explain how a fuse protects an electrical circuit. Why is the fuse wire connected in series rather than in parallel? **[4 marks]**

End of Question Paper